Dynamic Attribution-Gated Suppression for Cross-Dataset Fake News Detection

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Motivation

- Fake news detectors achieve ∼98% accuracy on training data but **collapse on unseen datasets**
- Root cause: **publisher bias** — models learn "Reuters = real", not "this content is deceptive"
- *The model reads the letterhead, not the article.*

Baseline gap: 0.965 in-dataset vs 0.751 cross-dataset
Δ = 0.214 — a 22-point performance collapse at deployment

- Publisher names, bylines, and editorial vocabulary act as **spurious shortcuts**
- Standard regularisation and data augmentation cannot remove these features

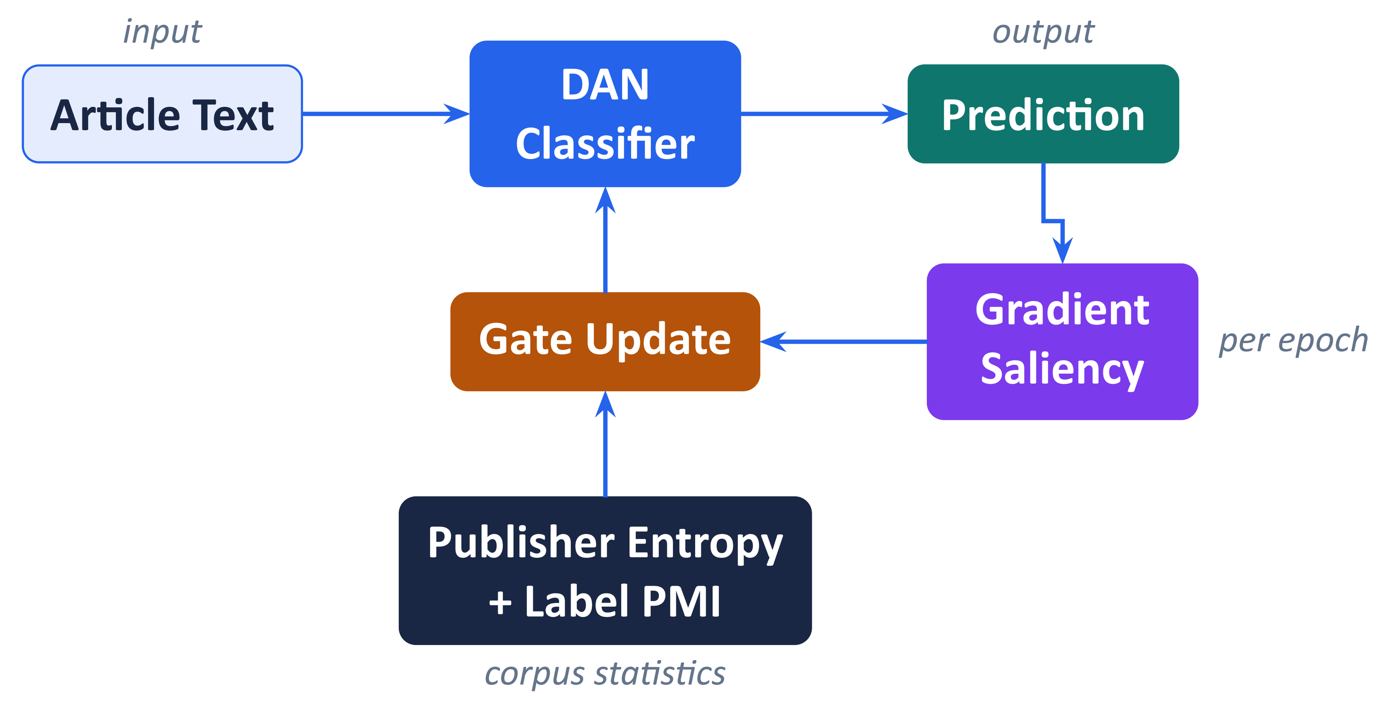
The Problem with Static Methods

- Existing debiasing: compute a suspicious token list **once before training**, apply forever
- Static masks suppress tokens the model isn't even using

Wrongly suppressed tokens:
`rohingya, myanmar, tariffs, lighthizer`
High publisher concentration → but *low model reliance*

- **Hard mask** damages cross-dataset performance by removing meaningful content alongside real fingerprints
- No feedback loop — mask cannot adapt as model representations evolve
- Statistics-only gating is **necessary but not sufficient** for suppression

Method — DAGS (Dynamic Attribution-Gated Suppression)



Closed loop — gate adapts every epoch unlike static methods

Deep Averaging Network

Embedding → mean-pool → MLP. Trained **from scratch** — no CNN, no fine-tuning.

Publisher Entropy $H(t)$

Low entropy = token in one publisher = potential fingerprint. Computed over the full corpus.

Gradient Saliency

After each epoch, measure which tokens the model is *actually relying on* via input-embedding gradients.

Label PMI

Pointwise mutual information between token and label. Captures class-predictive tokens beyond publisher signal.

$$\text{score}(t) = \alpha \cdot C_{\text{pub}} + \beta \cdot |\text{PMI}| + \gamma \cdot S_{\text{saliency}}$$

Suppress only tokens that are BOTH publisher-specific AND model-relied-upon

CLOSED LOOP → Gate adapts every epoch unlike static one-shot methods

Gate Analysis — The Smoking Gun

Initialization (statistics only)

Token	Entropy
`reuters`	$H = 0.000$
`rohingya`	$H = 0.088$
`myanmar`	$H = 0.091$
`tariffs`	$H = 0.102$
`lighthizer`	$H = 0.134$

All flagged by statistics alone

After Saliency Feedback

`reuters` **kept** ✓ model relies on it

28 geopolitical tokens
RELEASED — model not using them

New additions: actual bylines model was cheating with

Static methods would have wrongly suppressed 28 content tokens with geopolitical relevance. DAGS released them because gradient saliency

Results

Method	In-dataset	Cross-dataset	Gap
Baseline	$0.965_{\pm 0.001}$	$0.751_{\pm 0.005}$	$0.214_{\pm 0.006}$
Hard Mask	$0.961_{\pm 0.005}$	$0.747_{\pm 0.000}$	$0.214_{\pm 0.005}$
Soft Mask	$0.961_{\pm 0.002}$	$0.751_{\pm 0.005}$	$0.209_{\pm 0.004}$
DAGS (ours) ★	$\mathbf{0.965}_{\pm 0.002}$	$\mathbf{0.754}_{\pm 0.003}^{\dagger}$	$\mathbf{0.210}_{\pm 0.002}^{\ddagger}$

† Highest cross-dataset accuracy across all methods
‡ Lowest variance — most stable generalisation across seeds
★ Only method matching baseline in-dataset while improving out-of-distribution

Synthetic Proof of Concept

0.438 → **0.032**
Baseline gap → DAGS gap

92.6% gap reduction on synthetic data

Proves the mechanism works when bias is token-localised. Controlled experiment confirms saliency-gated suppression targets the right tokens.

Conclusion

- ★ **First closed-loop** token suppression using gradient saliency for fake news debiasing
- ↑ Achieves **highest cross-dataset accuracy** + **lowest prediction variance** across all methods
- ✓ Gate analysis proves static methods **over-suppress**: 28 meaningful tokens wrongly removed by statistics alone
- ◊ Real-world publisher bias is distributed across **editorial vocabulary** — motivates future phrase-level and embedding-space debiasing approaches
- ∘ 92.6% gap reduction on synthetic data confirms mechanism when bias is token-localised

`Code available at: https://github.com/koopticon/fake-news-debias.git`